VIJAYA VITTALA INSTITUTE OF TECHNOLOGY

#35/1, Dodda Gubbi Post, Hennur-Bagalur Road, Bengaluru-560077
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING IN AIML

Subject Name : MACHINE LEARNING II
Subject Code : BAI702

MODULE 1- QUESTION BANK

SL. NO	Questions	CO	RBT
01	Explain the fundamental design issues and approaches considered while developing a machine learning program for playing checker game	CO1	L2
02	Describe the key perspectives and issues in machine learning	CO1	L2
03	State and prove the Version space representation theorem. Clearly show that: i) Every hypothesis 'h' satisfying $(\exists s \in S)\ (\exists g \in G)\ (g \geq_g h \geq s)$ is a member of $VS_{H,D}$ ii) Every hypothesis in $VS_{H,D}$ satisfies this condition.	CO1	L3
4	Describe well-posed learning problems in machine learning with suitable examples	CO1	L2
05	Explain Enjoy Sport concept learning task	CO1	L2
06	Outline the step involved in candidate-elimination algorithm for computing version space and explain with an example	CO1	L3
07	What is a Learning System? Explain its components with a block diagram.	CO1	L2
08	Define Version Space. What does it represent?	CO1	L1
09	What are specific and general boundaries in version space?	CO1	L1

Faculty Name

Prof. Dhananjay K R

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MODULE 2- QUESTION BANK

SL. NO	Questions	CO	RBT
01	Explain general – to – specific beam search and LEARN – ONE – RULE algorithm	CO2	L2
02	Using a small relational dataset, explain the FOIL algorithm and apply the same to generate first – order rule.	CO2	L3
03	Outline the explanation – based learning algorithm PROLOG – EBG with help of Safe To Stack examples	CO2	L3
04	Describe how explanation based learning can be used to derive search control knowledge	CO2	L2
05	List the three different methods for using prior knowledge to alter the search performed by purely inductive methods.	CO2	L1
06	Explain Learning Sets of Rules using the Sequential Covering approach	CO2	L2
07	Describe CN2 or RIPPER algorithm for rule induction	CO2	L2
08	What is rule pruning? Why is it required?	CO2	L2
09	Explain Analytical Learning with respect to perfect domain theories	CO2	L2
10	Describe Explanation-Based Learning of search control knowledge.	CO2	L2
11	List advantages and limitations of EBL	CO2	L1

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MODULE 3- QUESTION BANK

SL. NO	Questions	CO	RBT
01	Explain the adaboost algorithm and how it sequentially improves weak classifiers and with weight update equations	CO3	L3
02	Outline the step of basic random forest training algorithm	CO3	L2
03	Explain the k-means algorithm with an example	CO3	L3
04	Explain ensemble learning and why combining multiple models often improves performance compared to a single model & it's motivation	CO3	L2
05	Explain the K – Means neural network algorithm	CO3	L2
06	Outline the steps of the mixture of experts algorithm	CO3	L2
07	Describe Bagging algorithm with a flowchart.	CO3	L2
08	What is out-of-bag error?	CO3	L1
09	Compare Single classifier vs Ensemble classifier.	CO3	L4
10	Describe Random Forest construction process.	CO3	L2
11	Explain k-means clustering and handling of noise.	CO3	L3
12	Explain the effect of initial centroid selection	CO3	L4

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MODULE 4- QUESTION BANK

SL. NO	Questions	CO	RBT
01	Describe the vector quantization with diagram that shows an interpretation of prototype vectors in two dimensions	CO3	L2
02	Illustrate the self organizing feature map algorithm	CO3	L3
03	Explain the following with respect to self organizing features map. I) Self – Organization II) Network dimensionality and boundary conditions	CO3	L2
04	Discuss the Following: I) The Box – Muller Scheme II) Monte Carlo Principle	CO4	L2
05	Describe Markov Chain Monte Carlo (MCMC) and Explain Metropolis Hastings algorithm used for MCMC	CO4	L3
06	Explain Vector Quantisation and its role in clustering	CO3	L2
07	Describe Self-Organizing Feature Map (SOFM) learning algorithm.	CO3	L2
08	Explain network dimensionality and boundary conditions in SOM	CO3	L2

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MODULE 5- QUESTION BANK

SL. NO	Questions	CO	RBT
01	Explain the Hidden Markov Models (HMMs) Forward algorithm and HMM Viterbi algorithm	CO4	L3
02	Illustrate the step of Kalman filter algorithm and particle filter algorithm	CO4	L3
03	Explain the Baum – Welch (forward-backward) algorithm	CO4	L3
04	Describe Bayesian Networks and approximate inference.	CO4	L2
05	Explain Markov Random Fields.	CO4	L2
06	Explain Bayesian Networks with a suitable example.	CO4	L2
07	Explain applications of Hidden Markov Models.	CO4	L2

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